

Introduction to Simulation-Based Inference

Paul Bürkner based on materials of Šimon Kucharský

Artificial Intelligence has two goals. First, AI is directed toward getting computers to be smart and do smart things so that human beings don't have to do them. And second, AI [...] is also directed at using computers to simulate human beings, so that we can find out how humans work.

Simon (1983, p. 27), van Rooij et al. (2024)

What are the values of the model parameters θ , given observed data x ?

Bayes' theorem

$$\begin{aligned} p(\theta \mid x) &= \frac{p(\theta, x)}{p(x)} \\ &= \frac{p(\theta) \times p(x \mid \theta)}{\int p(\theta) \times p(x \mid \theta) d\theta} \end{aligned}$$

$$p(x) = \int p(\theta) \times p(x \mid \theta) d\theta$$

- difficult to evaluate
- often intractable

- Approximate $p(\theta \mid x)$
 - Markov Chain Monte Carlo (MCMC): $\theta \propto p(\theta) \times p(x \mid \theta)$
- Obtain point estimates:
 - Maximum likelihood: $\hat{\theta} = \operatorname{argmax}_{\theta} p(x \mid \theta)$
 - Maximum aposteriori: $\hat{\theta} = \operatorname{argmax}_{\theta} p(\theta) \times p(x \mid \theta)$
- None of the methods require $p(x)$
- But all require evaluating $p(x \mid \theta)$

- “Likelihood-free”
- When we cannot even evaluate $p(x \mid \theta)$
- Approximate using sampling

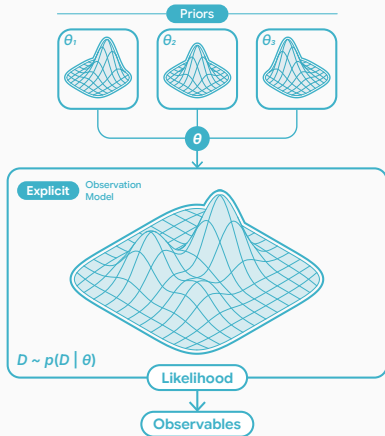
Examples

1. Surrogate Likelihood
2. Rejection Sampling
3. Approximate Bayesian Computation

Modern overview: Cranmer et al. (2020)

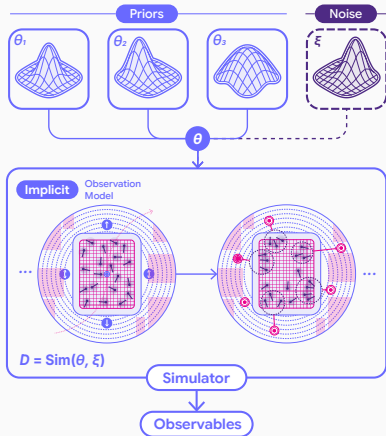
Likelihood-Based vs. Simulation-Based Inference

LIKELIHOOD-BASED MODELS



VS

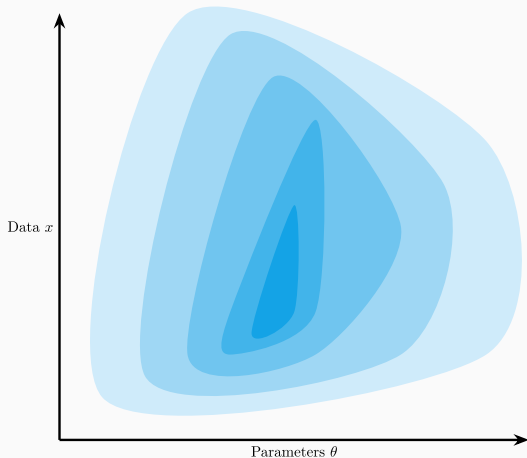
SIMULATION-BASED MODELS



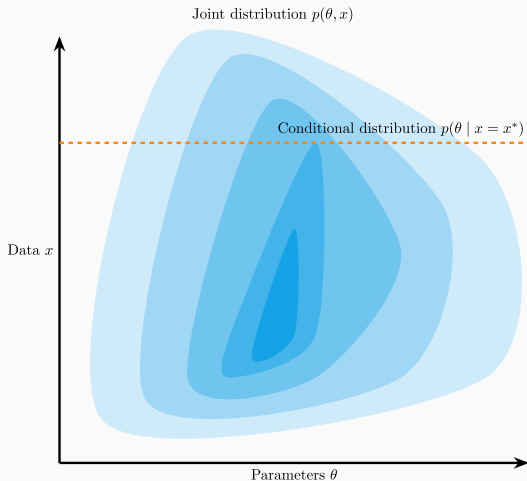
- For a given parameter value θ , simulate many samples of x
- Estimate the density $p(x \mid \theta)$ (e.g., kernel density estimation)
 - Used for approximate MLE/MAP, or within MCMC

$$p(\theta, x) = p(\theta)p(x \mid \theta)$$

Joint distribution $p(\theta, x)$



$$p(\theta, x) = p(\theta)p(x | \theta)$$



$$\theta^{(s)} \sim \text{Beta}(1, 1)$$

$$x^{(s)} \sim \text{Binomial}(\theta^{(s)}, 10)$$

$$\theta \mid x^{\text{obs}} = 7 \approx \text{Samples from } \theta^{(s)} \text{ where } x^{(s)} = 7$$

Listing 1 Python

```
prior = np.random.beta(1, 1, size=5000)
x = np.random.binomial(n=10, p=prior)

observed = 7
posterior = prior[x == observed]
```

Time for exercise `rejection-sampling`

Approximate Bayesian Computation (ABC)

- Generalization of rejection sampling
- Given a sampled parameter value θ , generate a data set x
- Compare the simulated data set to observed x^{obs}
- Retain the parameter if the data sets are not too dissimilar

$$\rho(s(x), s(x^{\text{obs}})) \leq \epsilon$$

- Curse of dimensionality
- Computationally expensive
- Handcrafted summary statistics

- Approximate using *generative neural networks*
 - Deep learning architecture that approximates a probability distribution

Neural likelihood estimation (NLE)

- Learn $p(x \mid \theta)$
- Surrogate likelihood

Neural posterior estimation (NPE)

- Learn $p(\theta \mid x)$
- Obtain posterior directly

- Generative neural networks
 - Produce a distribution $q(\theta \mid x)$
- Train them on *simulated* pairs $(\theta^{(s)}, x^{(s)}) \sim p(\theta, x)$
 - Learn $q(\theta \mid x) \approx p(\theta \mid x)$
- Once trained, can be used on observed data x^{obs}
 - $q(\theta \mid x^{\text{obs}}) \approx p(\theta \mid x^{\text{obs}})$

Pay the cost of inference upfront during training, receive benefits later

Training

- Train neural networks
- Simulated data and parameters
- Learn the mapping between data and parameters
- Slow, resource consuming process

Inference

- Apply pretrained networks
- Observed data
- Posterior distribution of parameters
- Fast, cheap process

Using deep learning generative neural networks to make Bayesian inference.

Advantages

- Fast inference
- Simulation based – intractable models

Disatvantages

- Need for training
- Simulation based – weaker guarantees

- Cranmer, K., Brehmer, J., & Louppe, G. (2020). The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117(48), 30055–30062.
- Simon, H. A. (1983). Why should machines learn? In *Machine learning* (pp. 25–37). Elsevier.
- van Rooij, I., Guest, O., Adolfs, F., Haan, R. de, Kolokolova, A., & Rich, P. (2024). Reclaiming AI as a theoretical tool for cognitive science. *Computational Brain & Behavior*, 7(4), 616–636.